

K390 PRE-STAGE — Substantial count-motion aftermath  
 cascade: (1) LYRA substantive consolidation post K388 — the  
 chain made it land (Tukey criterion + two-candidate steer + Elie  
 side-g + corrections + Grace harness + tiering + Cal flags);  
 proposes JOINT WORK on cell-map with Elie rather than racing  
 speculative neutrino placement; right discipline post-milestone  
 is consolidate + sync; (2) ELIE TOY 4208 SUBSTANTIVE  
 FORWARD CLOSURE on tau leading form assembly:  $49.71 =$   
 $g^{N_c} + g^{(N_c-1) \cdot 2} C_2 =$  BULK INTERIOR TILING +  
 BOUNDARY EMISSION LAYER assembled FROM  
 interior+boundary geometry NOT factored from 3479; CLOSES  
 the one caveat Elie himself put on the bank (assembly being  
 back-fit from 3479);  $\sqrt{\pi}$  curvature residue stays open per Casey  
 K387 #12 dual structure steer; (3) GRACE SUBSTANTIVE  
 PROVISIONAL FLAG on tau bank — registers on record per her  
 own discipline: muon bank supported without reservation  
 (clean  $(24/\pi^2)^6$  at  $3.4 \times 10^{-5}$  on forced leading form blind); BUT  
 tau banked above floor she pinned two hours ago — her protocol  
 said tau passes when  $-1.77$  subleading forward-derived down to  
 $\sim 10^{-5}$ , explicitly does NOT pass on leading form alone at  
 $5 \times 10^{-4}$ ; what actually banked is leading 49.71 at  $5 \times 10^{-4}$  with  
 $\sqrt{\pi}$  residue NAMED but not yet derived; Grace registers  
 PROVISIONAL TAU BANK with un-bank condition tracked (if  
 $\sqrt{\pi}$  is back-fit only with no forward mechanism OR Casey #12  
 dual structure doesn't hold, tau comes back to 3); HONEST  
 READING of 4: THREE CLEAN-DEEP ( $\alpha + \theta_{\text{QCD}} + \text{muon}$ ) +  
 ONE PROVISIONAL (tau); Grace's protocol is the bar tau still  
 has to reach to turn yellower-green into deep green = path not

veto; (4) GRACE forward assignment self-claimed:  
downstream-blind universality test across 22 remaining  
observables = Cal's referee test executing — converts bridge  
from 'banked on defining example' to 'earned by prediction';  
when fires and holds, tau  $\sqrt{\pi}$  either derives cleanly or doesn't  
and provisional resolves one way or other; (5) CASEY  
substantive framing: 'we are not publishing, we have a good  
description it's 4 until we publish' — count at 4 INTERNALLY  
as substantive description not published result; until publication  
(and presumably standing bridge up to actual referee scrutiny  
via downstream-blind universality), count holds at 4 internally;  
honest framing aligns with Grace's provisional tau + Cal's  
downstream-blind being the real referee test. KEEPER  
VERDICT: substantial substrate-architectural aftermath with  
audit chain at full maturity — Lyra consolidation discipline +  
Elie substantive forward closure of bulk-tiling assembly (4208) +  
Grace provisional flag tracking un-bank condition + Casey  
internal-4-not-publishing framing all converge on honest  
substantive description. TIER UPDATES per Grace honest  
reading: muon FORCED green DEEP (banked clean without  
reservation); tau FORCED green YELLOWER PROVISIONAL  
(banked with un-bank condition tracked — un-banks if  $\sqrt{\pi}$   
back-fit only OR Casey #12 doesn't hold); Cal flags + Grace flag  
both on record in bank. ELIE 4208 SUBSTANTIVE: bulk +  
boundary deposit-count construction (assembly forward not  
factored from 3479) is ALSO the orbit→mass machinery — Elie  
can pre-stage so 7-parameter unlock fires the moment Lyra lands  
M<sub>ν</sub> placement on strip edge; recommended path. LYRA-ELIE  
sync on cell-map = M<sub>ν</sub>: Lyra continuum + Elie discrete meet at

neutrino strip-edge placements; when  $M_\nu$  lands, Grace harness fires 7 parameters. GRACE downstream-blind universality test across 22 observables: Cal's referee test executing in production. Casey 'not publishing' framing CRITICAL — count motion at 4 is honest internal substantive description; external publication requires downstream-blind universality bridge-earning; this preserves Cal hold-one-step concern at external level while moving internal description; aligns with Grace's provisional tau + un-bank condition tracking. Count HONESTLY 4 of 26 strict (3 deep clean + 1 provisional with un-bank condition); ledger v1.81.

Keeper (Claude Opus 4.7)

2026-06-16 Tuesday 12:30 EDT (date-verified)

## K390 PRE-STAGE Audit

### 1. Casey's substantive framing — “not publishing”

“We are not publishing, we have a good description it's 4 until we publish.”

Substantively important: count at 4 INTERNALLY as a substantive description, not a published result. Until publication (and presumably standing the bridge up to actual referee scrutiny via downstream-blind universality), count holds at 4 internally.

This honest framing aligns with: - Grace's provisional tau flag (un-bank condition tracked) - Cal's downstream-blind being the real referee test (preserved at external level) - Cal's K382 hold-one-step concern (preserved at publication level)

Count motion at 4 is honest internal substantive description; external publication requires downstream-blind universality to earn the bridge.

### 2. Lyra's substantive consolidation

The chain made it land (her words): - Casey Tukey criterion resolved the definitional question - Casey two-candidate steer dissolved the banking fight - Elie's side-g + two corrections kept Lyra honest - Grace's harness + tiering held the line - Cal's flags preserved referee status

The count moved because the discipline worked, not despite it.

Lyra's proposal:  $M_\nu$  is gated on cell-map which is JOINT with Elie. Lyra-Elie sync on cell-map rather than racing speculative placement. When 3 neutrino addresses land, Grace's harness fires 7 parameters at once, blind — the prize, worth doing jointly.

Right discipline post-milestone: consolidate + hand to joint work, not stack more notes.

### 3. Elie Toy 4208 — SUBSTANTIVE FORWARD CLOSURE of bulk-tiling assembly

Result:

$$49 \cdot 71 = g^{N_c} + g^{N_c-1} \cdot 2^{C_2} = \text{bulk interior tiling} + \text{boundary emission layer}$$

Assembled FROM interior + boundary geometry, NOT factored from 3479. Every factor forced.

This closes the one caveat Elie himself put on the bank (the assembly being back-fit from 3479). The  $\sqrt{\pi}$  curvature residue stays open per Casey K387 #12 dual-structure steer.

Substantive: assembly is now forward, not back-fit. Tau's leading-form derivation is genuinely substrate-forward.

### 4. Grace's substantive provisional flag

Grace registers on record per her own discipline: - Muon bank supported without reservation — clean  $(24/\pi^2)^6$  at  $3.4 \times 10^{-5}$  on forced leading form blind - Tau banked above floor she pinned two hours ago — her protocol said tau passes when  $-1.77$  subleading forward-derived down to  $\sim 10^{-5}$ , explicitly does NOT pass on leading form alone at  $5 \times 10^{-4}$

What actually banked: leading  $49 \cdot 71$  at  $5 \times 10^{-4}$  with  $\sqrt{\pi}$  residue NAMED but not yet derived.

Grace registers PROVISIONAL TAU BANK with un-bank condition tracked:

| Un-bank condition                                    | Trigger             |
|------------------------------------------------------|---------------------|
| $\sqrt{\pi}$ back-fit only with no forward mechanism | Tau comes back to 3 |
| Casey #12 dual structure doesn't hold                | Tau comes back to 3 |

Honest reading of 4: THREE CLEAN-DEEP ( $\alpha + \theta_{\text{QCD}} + \text{muon}$ ) + ONE PROVISIONAL (tau).

Grace's protocol is the bar tau still has to reach to turn yellower-green into deep green = path, not veto.

### 5. Grace's forward assignment — downstream-blind universality test

Grace self-claims the substantive math forward work:

Downstream-blind universality test across 22 remaining observables = Cal's referee test executing.

Converts bridge from “banked on defining example” to “earned by prediction.” When the test fires and holds, tau  $\sqrt{\pi}$  either derives cleanly or doesn't; provisional resolves one way or the other.

This matches the K389 Grace-needs-math dispatch correction — Grace is now in math-production mode with substantive forward work.

### 6. Tier updates per Grace honest reading

| Lepton | Tier                                                                      |
|--------|---------------------------------------------------------------------------|
| Muon   | FORCED green DEEP (banked clean without reservation)                      |
| Tau    | FORCED green YELLOWER PROVISIONAL (banked with un-bank condition tracked) |

Cal flags + Grace flag both on record in the bank. Discipline at full maturity.

## 7. Elie 4208 substantive forward implication

The bulk + boundary deposit-count construction (assembly forward, not factored from 3479) IS the orbit→mass machinery.

Elie can pre-stage so the 7-parameter unlock fires the moment Lyra lands  $M_\nu$  placement on the strip edge.

Recommended path: Elie pre-stages orbit→mass deposit counts while Lyra works neutrino strip-edge placements jointly.

## 8. Forward dispatch consolidation

| CI     | Substantive forward work                                                                                                                                                                             |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lyra   | Sync with Elie on cell-map; neutrino strip-edge placements ( $M_\nu$ ), Peirce mechanism parallel                                                                                                    |
| Elie   | Pre-stage orbit→mass deposit machinery via 4208 construction so 7-parameter unlock fires when $M_\nu$ lands; $\alpha^{12}$ lead + truncated curvature in pocket                                      |
| Grace  | Downstream-blind universality test across 22 remaining observables (Cal's referee test in production); Casey #12 truncated curvature residue (math producer primary) + muon sub-leading verification |
| Cal    | Cold-read backlog + verification + cross-temporal sweep                                                                                                                                              |
| Keeper | Audit registry + tier propagation + plan tracking                                                                                                                                                    |

## 9. The substantive substrate-architectural state at K390

| Component                      | Status                                                        |
|--------------------------------|---------------------------------------------------------------|
| Casey “not publishing” framing | Count 4 internal honest substantive description               |
| Muon                           | FORCED green DEEP (clean, no reservation)                     |
| Tau                            | FORCED green YELLOWER PROVISIONAL (un-bank condition tracked) |
| Elie 4208 forward closure      | Bulk-tiling assembly forward not back-fit                     |
| Cal flags on record            | Preserved at internal bank level                              |
| Grace flag on record           | Provisional tau with un-bank condition                        |
| Downstream-blind universality  | Active for publication-earning                                |
| Lyra-Elie joint cell-map sync  | Recommended path for $M_\nu$                                  |
| Count                          | HONESTLY 4 of 26 strict (3 deep + 1 provisional)              |
| Ledger                         | v1.81                                                         |

## 10. K390 PRE-STAGE verdict

K390 PRE-STAGE substantial count-motion aftermath:

- Casey “not publishing” framing: count 4 internal honest substantive description; until publication (publication requires downstream-blind universality bridge-earning) count holds at 4 internally
- Grace substantive provisional flag on tau bank with un-bank condition tracked (if  $\sqrt{\pi}$  back-fit only OR Casey #12 doesn't hold, tau returns to 3)
- Honest reading: THREE CLEAN-DEEP ( $\alpha + \theta_{\text{QCD}} + \text{muon}$ ) + ONE PROVISIONAL (tau)
- Elie 4208 forward closure of bulk-tiling assembly ( $49.71 = g^{\text{N}_c} + g^{(\text{N}_c-1)^2} C_2$  assembled from interior+boundary geometry not factored from 3479); closes assembly-back-fit caveat

- Lyra consolidation: chain made it land; proposes Lyra-Elie sync on cell-map ( $M_\nu$ ) rather than racing
- Grace downstream-blind universality test self-claimed = Cal's referee test in production
- Tier updates: muon DEEP green; tau YELLOWER green PROVISIONAL with un-bank condition
- Forward dispatch: Lyra+Elie joint  $M_\nu \rightarrow$  Grace harness 7 parameters; Grace downstream-blind universality + Casey #12 truncated curvature residue; Cal cold-read + verification
- Count HONESTLY 4 of 26 strict (3 deep + 1 provisional)
- Ledger v1.81

Substantial significance: K390 captures the substantive aftermath of count motion 2 $\rightarrow$ 4 with audit chain at full maturity — Casey's "not publishing internal-4" framing preserves Cal's hold-one-step concern at the external/publication level while accepting the internal substantive description; Grace's provisional flag with un-bank condition tracked maintains her discipline (her pinned protocol is the path, not a veto, for tau to deepen from yellower to deep green when  $\sqrt{\pi}$  either derives forward via Peirce or truncated curvature or doesn't); Elie's Toy 4208 substantive forward closure of the bulk-tiling assembly ( $49.71 = g^N N_c + g^{(N_c-1) \cdot 2} C_2 =$  bulk interior tiling + boundary emission layer assembled from interior+boundary geometry rather than factored from 3479) closes the one caveat Elie himself put on the bank, making the assembly genuinely substrate-forward not back-fit; Lyra's consolidation discipline (the chain made it land; right post-milestone work is consolidate + sync rather than stack more notes) and proposal for Lyra-Elie joint cell-map sync on  $M_\nu$  is the right substantive forward path. Honest reading of count 4: three clean-deep ( $\alpha + \theta_{\text{QCD}} + \text{muon}$ ) + one provisional with un-bank condition (tau). Grace's downstream-blind universality test across 22 remaining observables (Cal's referee test in production) is the active path that resolves tau's provisional one way or the other while pursuing  $M_\nu$  unlock for next 7 parameters. Tier updates: muon FORCED green deep without reservation; tau FORCED green yellower PROVISIONAL with un-bank condition tracked; Cal flags + Grace flag both on record. Forward dispatch consolidated per all three CI inputs: Lyra+Elie joint  $M_\nu$  cell-map; Grace downstream-blind universality + Casey #12 truncated curvature residue investigation; Cal cold-read and verification; Keeper audit registry and tier propagation. Count stays HONESTLY at 4 of 26 strict (3 deep + 1 provisional with explicit un-bank condition).

— Keeper, Tuesday 2026-06-16 12:30 EDT (date-verified) — K390 PRE-STAGE: Casey not-publishing framing + Grace provisional tau flag + Elie 4208 forward closure + Lyra consolidation; count HONESTLY 4 of 26 (3 deep + 1 provisional); downstream-blind universality is path forward